xMCF v3.1.1
ISO/PAS 8329

Description of **mechanical** connections and joints in structural systems

- Introduction
  - Project Recap
    - Status

ISO/PAS 8329 xMCF WG meeting

2025-10-15

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energizing great minds

Agenda

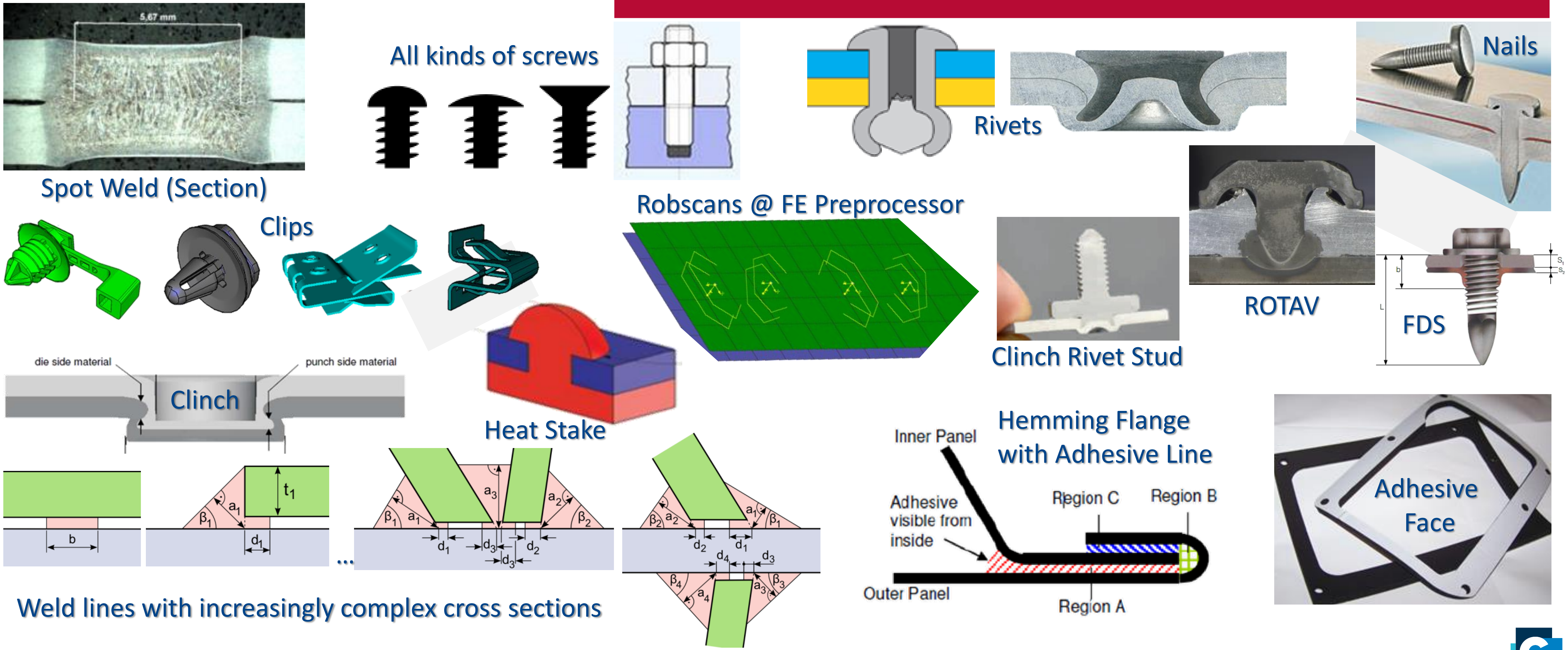
1	Joining Technologies & Processes are challenging …
2	χMCF – the Enabler for Smooth Processes
3	χMCF Use Cases
4	History, Status, Outlook & Supporting Organizations
5	Status of ISO Standardization
6	Federated use of STEP AP242 with χMCF optional
7	Technical Details of χMCF optional

Joining Technologies & Processes are challenging …

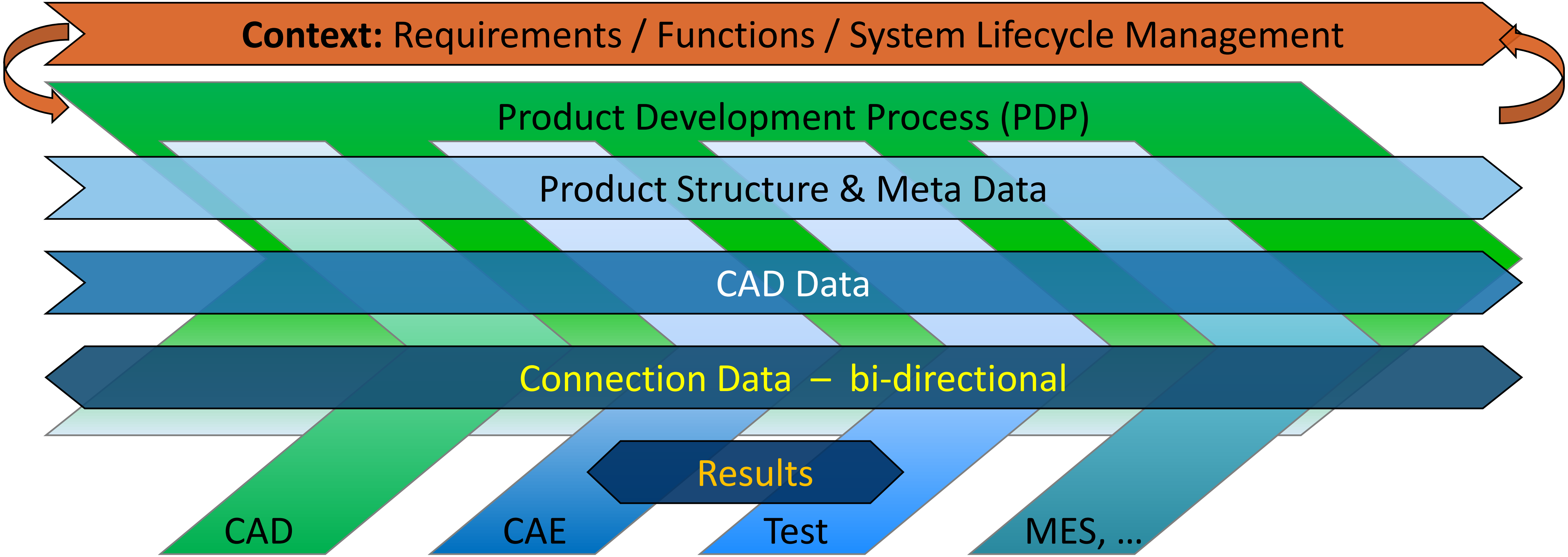
Challenges wrt. Connection Information

Big Variety & Complexity

- > 60 known connection techniques.
- Up to 25 quality criteria per connection technique.



Challenges wrt. Product Development Process



Connections are the mandatory ***3rd cross-sectional data track*** of PDP!

What is so Special about Connection Data?

Connections differ from product structure, meta data and CAD data, since e. g.:

function dominates over their shape

need more *PLM upstream data propagation* than CAD

work needs different *tools & plugins*, special *process steps* and expert *knowledge*

belong to *inner nodes* of product tree, not to its leaves

data size is much smaller than CAD data

CAD and connections *complement each other*. Each is *useless without the other*.

What are the Frequent Problems?

- Every OEM creates *own CATIA/NX macros* or buys *proprietary software*.
- Common suppliers *need to be familiar with all those* tools.
- Data exchange along process chain needs *additional tools*, frequently “home-brewed”.
  ⇒ Expensive and error-prone.
- However, in reality only *few techniques* are supported with only a *fraction of their data*.
- Inventing new techniques or adding new parameters
  results in *excessive costs* and *process threats*.
- Changing software vendors implies *high investments*.

⇒ Resulting “*vendor lock-in-effect*” impedes competition
and hence *hinders progress*.

χMCF – the Enabler for Smooth Processes

χMCF: *The* Standard for Connection Information

VDA | FAT Forschungsvereinigung Automobiltechnik | AK 25 Fügetechnik | χMCF WG defines and maintains χMCF.

XML-based χMCF meets all “usual” requirements to a standard **(incl. long-time ability!)**, plus:

- All connection types & techniques can be represented.
  – Unambiguous, completely, exact, and to the current design maturity within the PDP.
- All PLM processes are supported – CAD, CAE, CAT, Manufacturing Planning & Execution, including special sub processes, e. g.:
  - Durability simulation,
  - Robot programming,
  - Supplier integration, …
- One χMCF-file contains either data of one *assembly*, one *car* or all *variants* of a series.
  ⇒ χMCF meets any kind of *OEM specific process* design.
- χMCF allows imbedding *custom data* specific to *OEM, process* and *tool*.

⇒ All existing proprietary formats can be replaced sustainably. χMCF is like JT, but for connection data.

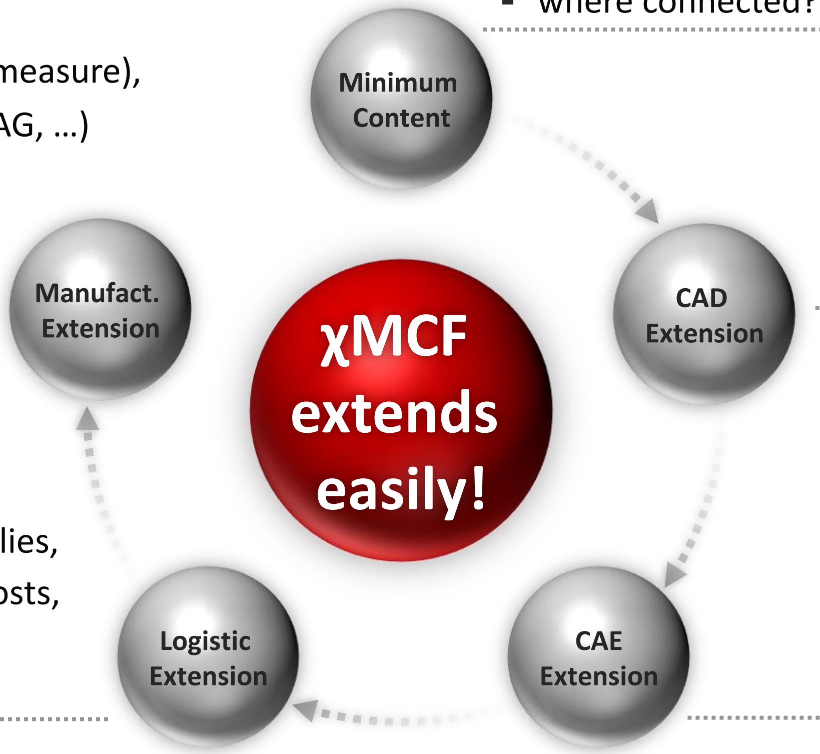
- VDA: German Association of the Automotive Industry
- FAT: Research Association for Automotive Technology; department of VDA
- AK 25: Working Group 25; focus on joining technologies

Download χMCF 3.1 Standard at VDA web site: https://www.vda.de/en/news/publications/publication/xmcf---a-standard-for-describing-connections---joints-in-mechanical-structures--version-3.1-

χMCF Accumulates Data along the Process Chain

- What is (part codes)
- how and
- where connected?

- Production quality (e. g. A-measure),
- Technology (Laser, MIG/MAG, …)
- Screw torque and angle,
- Die of self piercing rivet,
- …



- Part code of screw/rivet,
- Direction of laser or screw,
- …

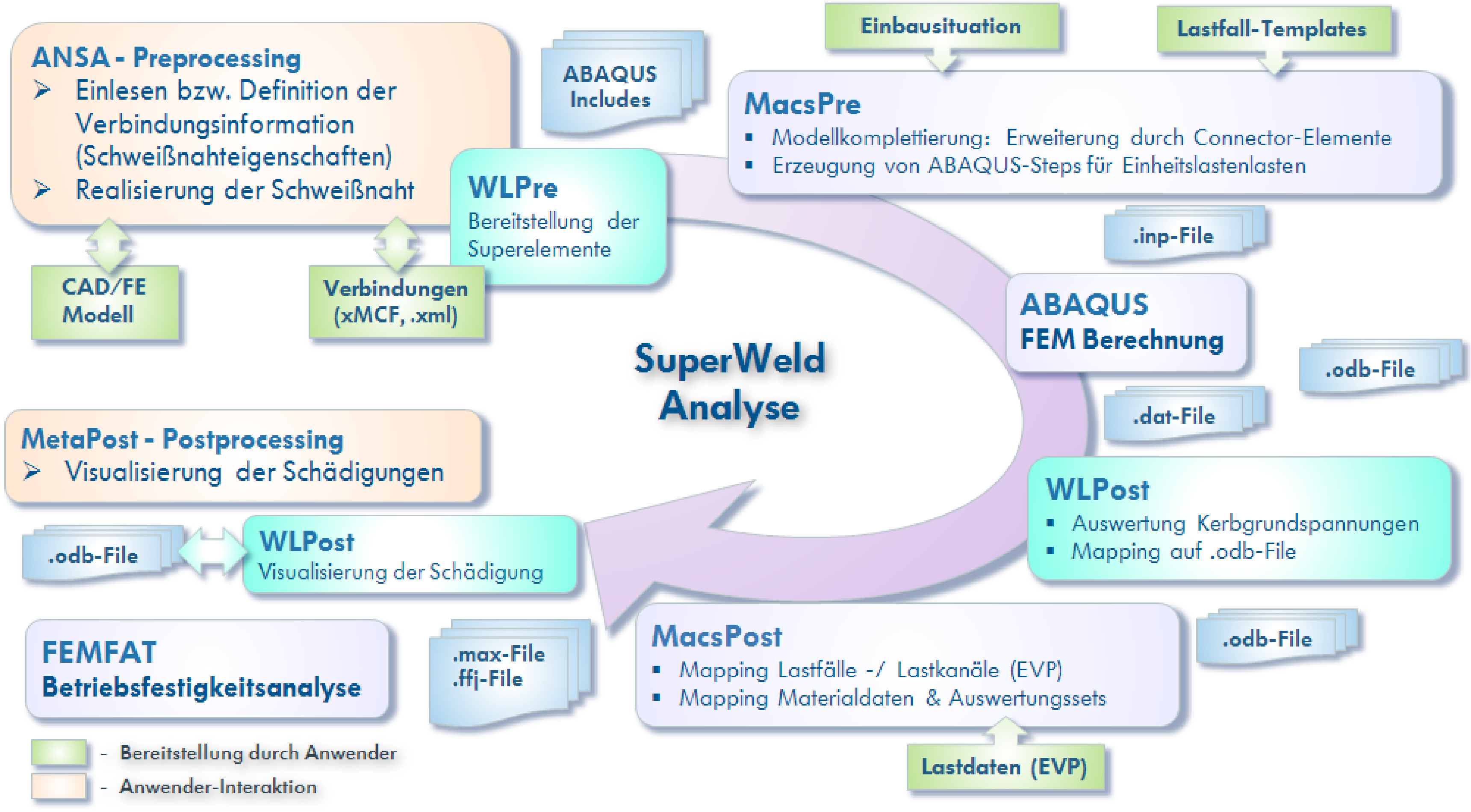
- PIDs of connected and connecting elements,
- Heat Affected Zones,
- Solver specific discretization parameters, e. g. hexahedra size, beam cross section,
- …

- Weight of screw,
- Shipping time for supplies,
- Unit costs, assembly costs,
- Reusability,
- …

χMCF Use Case
CAE Integration

Use Case 0: CAE Integration

Variant of a fatigue process at Volkswagen, 2014

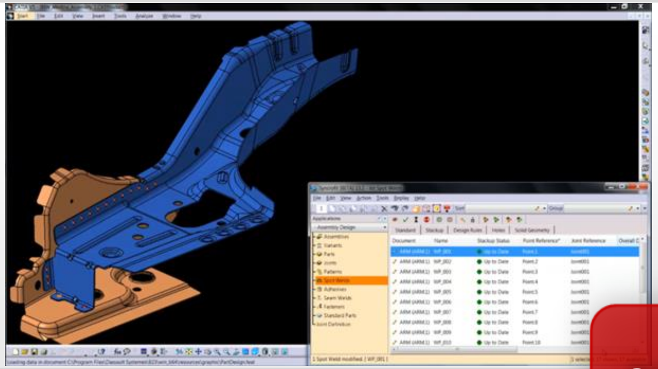


Many different tools.
Many different formats.

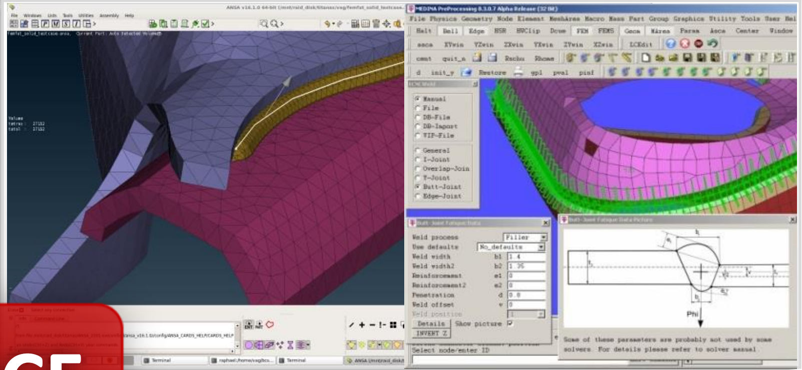
Source of image: Dr. Tröndle, Volkswagen AG, 2014

χMCF-based Tool-Integration (as of 2017)

CAD e.g. SyncroFIT for CATIA & NX

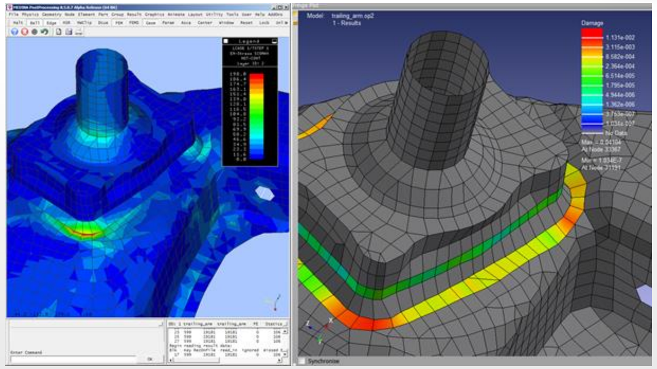


CAE Pre e.g. ANSA, MEDINA

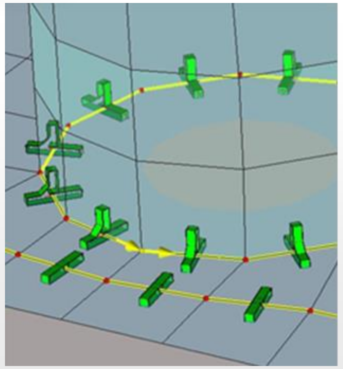


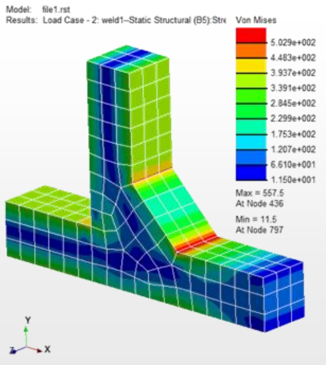
χMCF

CAE Post e.g. MEDINA, nCode



CAE Solve e.g. LMS, nCode, FEMFAT





Strategy & Example χMCF Use Cases at **BMW**

Use Case @ BMW (2018): χMCF Data Exchange between PDM & **Production Planning**

SAP IPPE

- Product structure data
  - metadata (PPXML)

PRISMA (TDM)

- Part information
  - metadata
  - geometrical data (JT)

VIP (Verbindungstechnisches Informations- und Produktionssystem)

- Manufacturing features data
  - metadata (**χMCF 3.0**)
  - geometrical data (JT)

Teamcenter Manufacturing

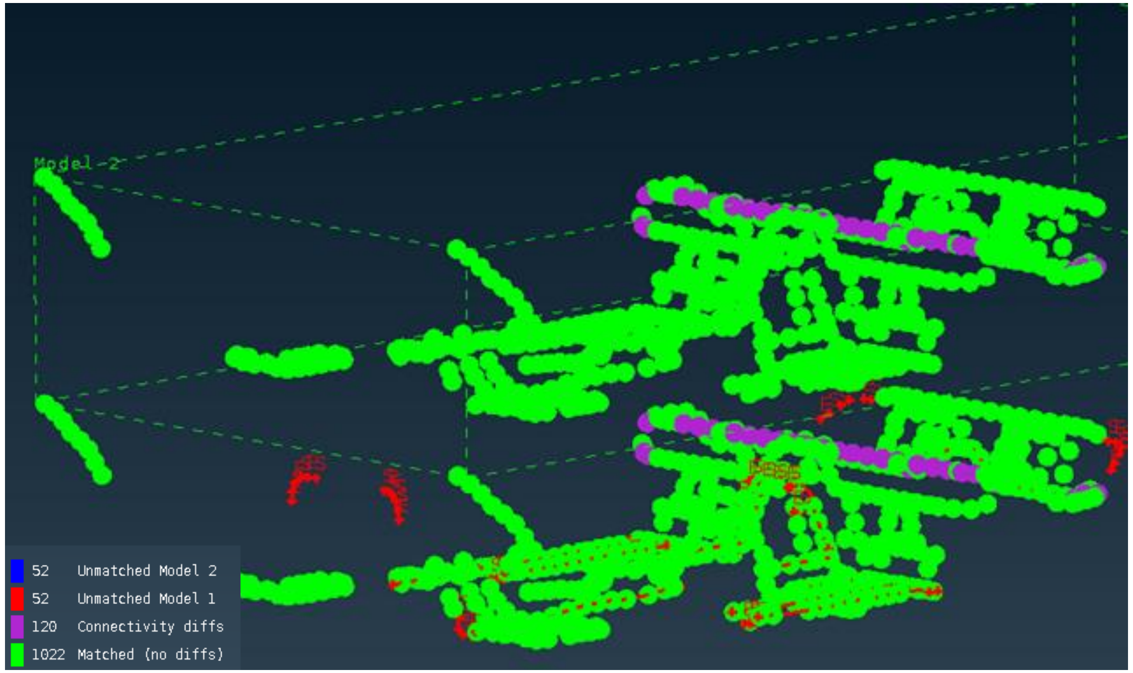
- EBOM
- BodyBOM
- BOP
- Plant BOM

BMW χMCF Strategy

- **Production Planning** (Teamcenter Manufacturing) was addressed 2018.

- **CAE** was addressed 2019. Data for CAE contains technology parameters, e.g. weld shape.

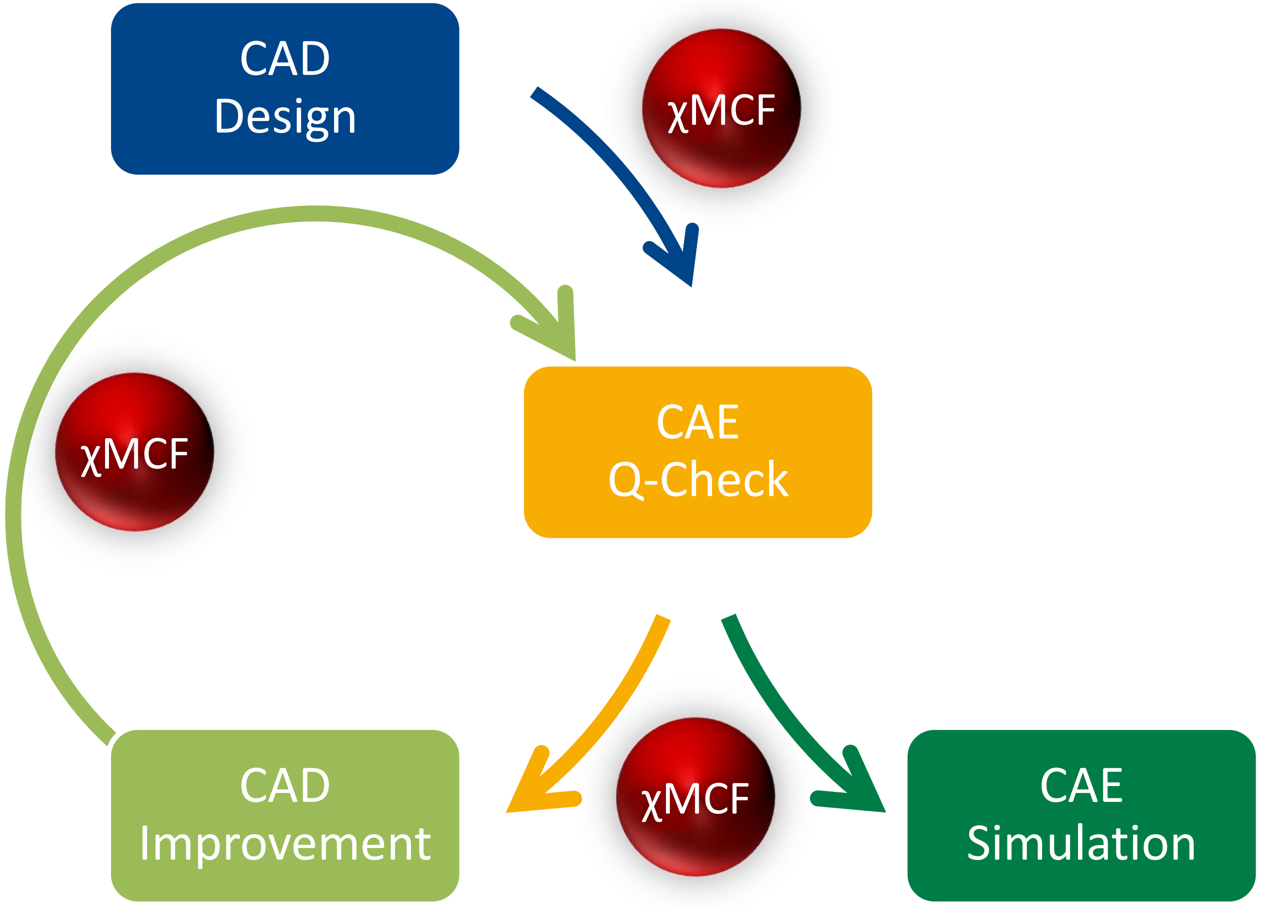
FEM preprocessor acted as a *verification tool*, comparing VIP & χMCF data during prototype phase:



χMCF Example Use Case at **Volkswagen**

Use Case @ at Volkswagen (2019): **Quality**-Gate between CAD and CAE – Enabled by χMCF

- Frequently, a complete digital vehicle is assembled *in CAE for the first time* in product development process.
- Using χMCF, connection data can be provided to CAE in the most *automated and low-error fashion*.
- Powerful features of FEM a preprocessor allow for *automated, fast and reliable quality checks*.
- Custom scripts provide *custom error categorization*.
- Via χMCF, categorized quality issues can be *sent back to design*.
- Categories allow CAD to fix the issues by a defined and *plannable process*.



Use Cases of prostep ivip’s PDM-IF:

χMCF and STEP AP 242 federated

- Dassault Aviation
- BMW

See Mr. Hirel's presentation on this topic!

optional

χMCF – A Standard with History and Broad Support

Well known brands beyond one roof at

VDA FAT | Forschungsvereinigung Automobiltechnik AK 25 | Fügetechnik

2001:
MCF V 1.0
(Ford)

2006:
χMCF
1st prop.

2011:
χMCF
V 1.0
V 1.1

2014:
χMCF
V 2.0

2015:
χMCF
V 2.1

2016:
χMCF
V 3.0

2020:
χMCF
V 3.1

2021:
χMCF
V 3.1.1
↦ ISO 8329
in 2024

Typical WG meeting agenda:
- Status of ISO standardization
- Software vendor's reports
- Project backlog

Open for new members!

… will be continued e.g. by:
- Catalogs for standard parts
- New innovative techniques
- …

(Progress transparent on the web.)

optional

Summary

- Connection processes are *rich and manifold* – so are the data.
- χMCF is *the* powerful and mature standard for piping connection data forward and backward through the product development process.
- It is able to *bridge any gap* between process steps or tools.
  – As has been shown by applications at BMW & Volkswagen.
- Many *important tools already support* χMCF – more to come.

⇒ AK 25 provides support when *optimizing processes to benefit* from χMCF.

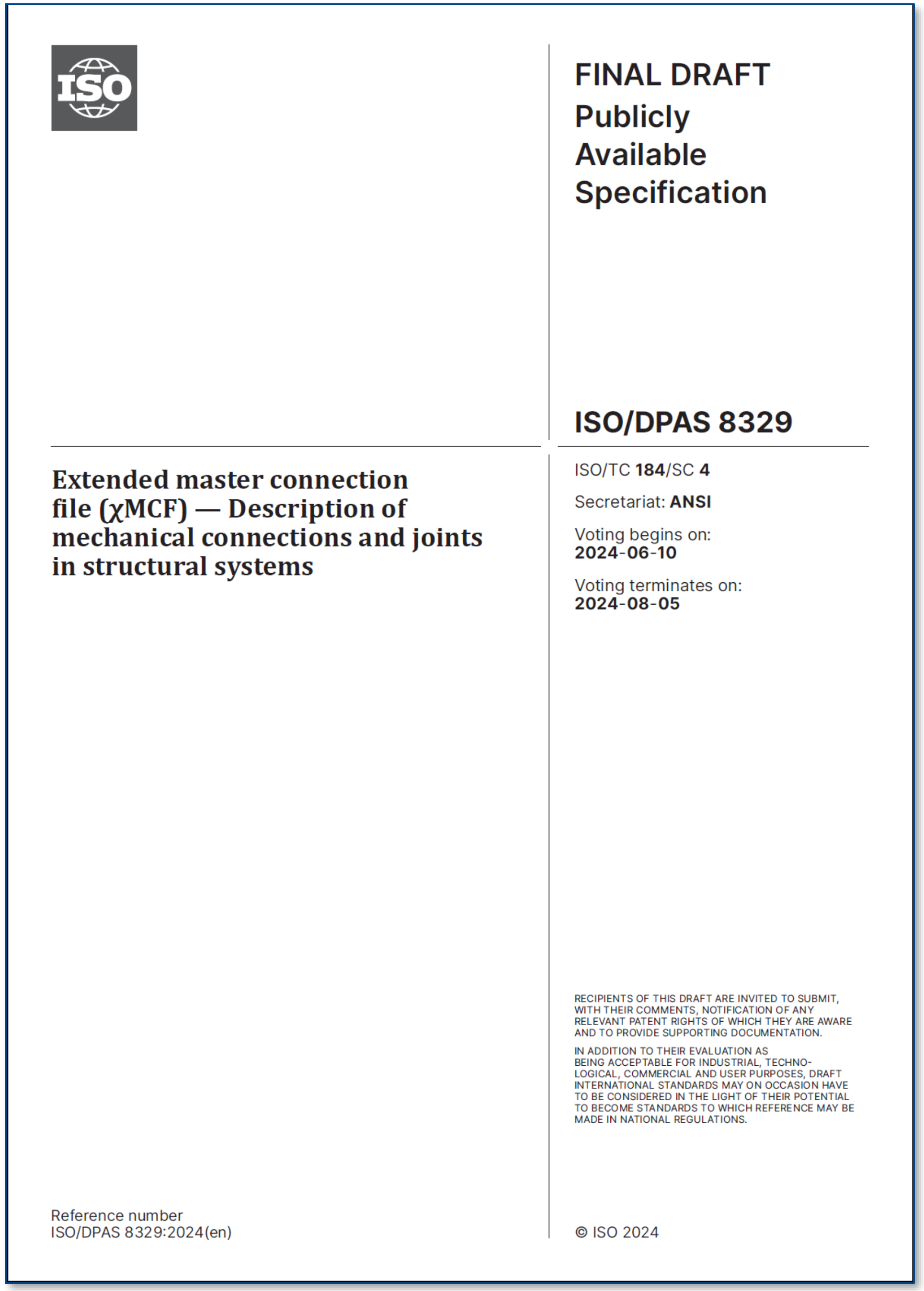
⇒ More *demand placed at software vendors* will lead to even wider support of χMCF.

Current Status of Standardization

- ISO/PAS 8329 describes χMCF 3.1.1,
  which contains minor improvements to 3.1.
- “PAS” stands for “Publicly Available Specification”.
- ISO Directive[1]: “A PAS shall remain valid for … 3 years.
  The validity may be extended for a single period up to a maximum of 3 years. …
  At the end of the validity period, the PAS shall be **transformed** with or without change into another type of normative document or shall be **automatically withdrawn**.”
- ISO Directive[2]: Final approval requires
  ≥ ⅔ pro of the participating members of ISO/TC 184[3]
  and **≤ ¼ contra** of all members.

⇒ We have a **challenge** here.

[1] See ISO regulation https://www.iso.org/sites/directives/current/consolidated/index.html#a_3_2_3.
[2] See ISO regulation https://www.iso.org/sites/directives/current/consolidated/index.html#a_2_7_3.
[3] See https://www.iso.org/committee/54110.html?view=participation


FINAL DRAFT
Publicly Available Specification

ISO/DPAS 8329

Extended master connection file (χMCF) — Description of mechanical connections and joints in structural systems

ISO/TC **184**/SC **4**

Secretariat: **ANSI**

Voting begins on:
2024-06-10

Voting terminates on:
2024-08-05

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IN ADDITION TO THEIR EVALUATION AS BEING ACCEPTABLE FOR INDUSTRIAL, TECHNO-LOGICAL, COMMERCIAL AND USER PURPOSES, DRAFT INTERNATIONAL STANDARDS MAY ON OCCASION HAVE TO BE CONSIDERED IN THE LIGHT OF THEIR POTENTIAL TO BECOME STANDARDS TO WHICH REFERENCE MAY BE MADE IN NATIONAL REGULATIONS.

Reference number
ISO/DPAS 8329:2024(en)

© ISO 2024

optional

Some selected milestones and events of χMCF WG & ISO Project

- 2021-10-21 WG releases χMCF 3.1.1. **Kickoff for ISO-standardization** project.
- 2022-02-03 WG welcomes Airbus & Volvo
- 2022-05-04 WG gets new management triple (Fox, Franke, Schmidt)
- Since 2023 ongoing exchange of WG with prostep ivip PDM-IF
- 2024-04-22 WG presents χMCF in joint WebCo of STEP AP242, LOTAR, MBxIF, PDMIF & EWISIF
- 2024-08-06 **ISO publishes** ISO/PAS 8329:2024 Extended master connection file (χMCF)
- 2025-02-17 “Common Use Cases” for χMCF discussed in PDM-IF MBx-IF Leadership Meeting
- 2025-03-20 Dassault Aviation requests new connection types (just as an example)
- 2025-05-13 PDM-IF (BMW & T-Systems) refers to χMCF at prostep ivip Symposium 2025
- 2025-09-24 χMCF was presented at VDA FAT AK25 Autumn Meeting
- 2025-10-16 χMCF will be presented at prostep ivip SSB

ISO standardization took almost 3 yrs.

optional

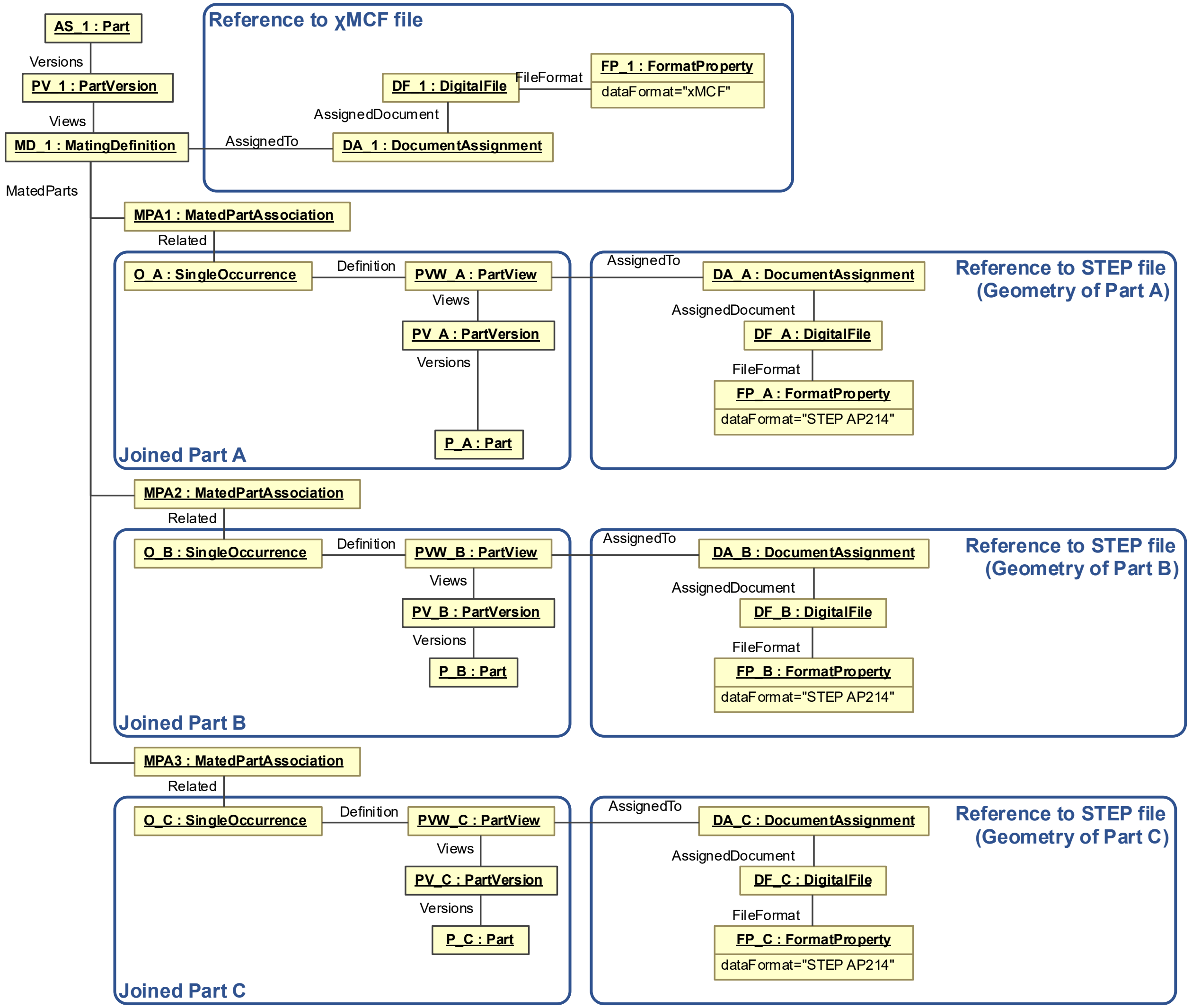
Federated use of STEP AP242 with χMCF

See Mr. Hirel's presentation on this topic!

Federating the Standards

Approach:

- Include *basic* mating information in AP 242 (existing capabilities)
- with linking to χMCF, where χMCF has a high level of details of the connection.



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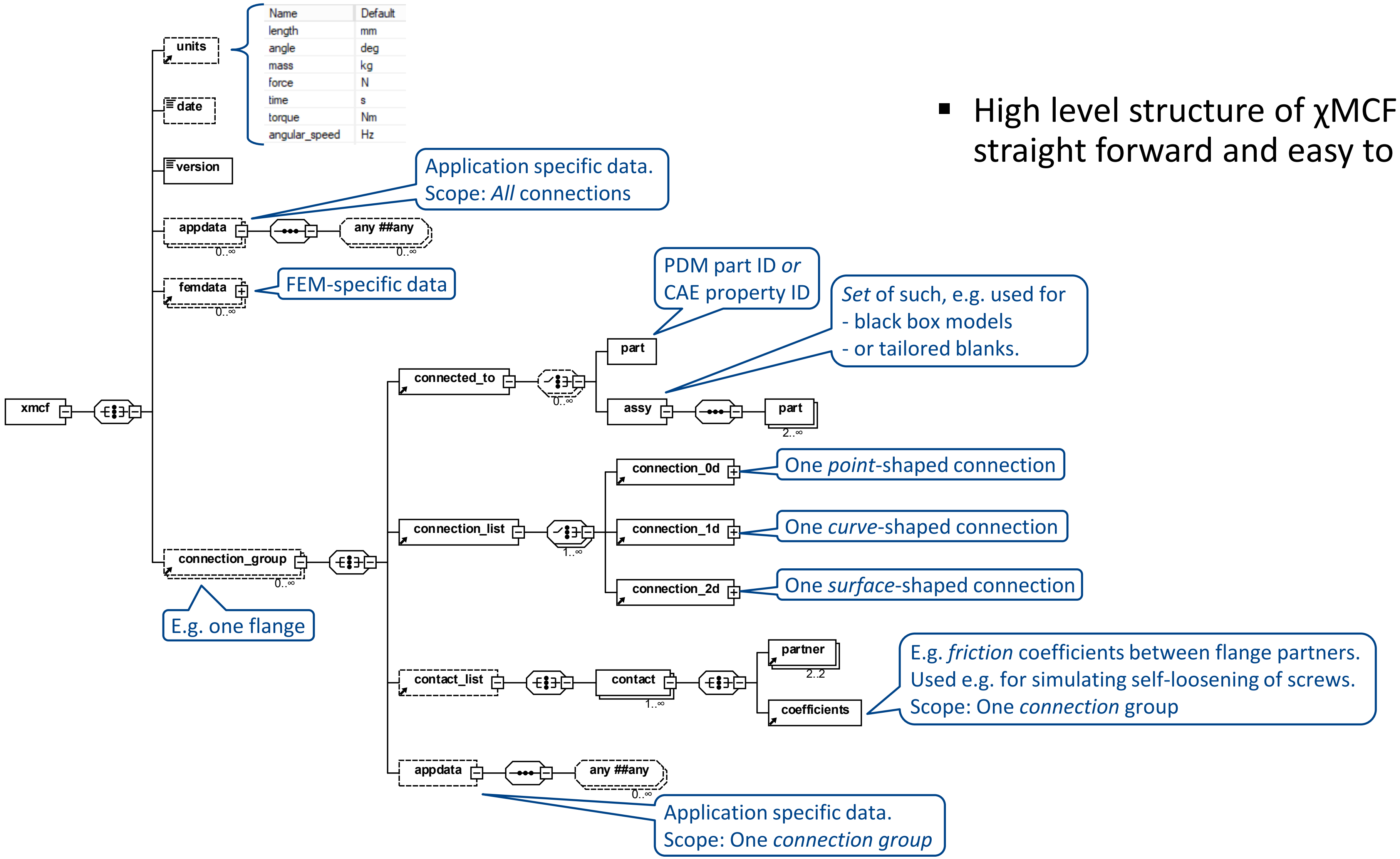
Technical Details of χMCF

optional

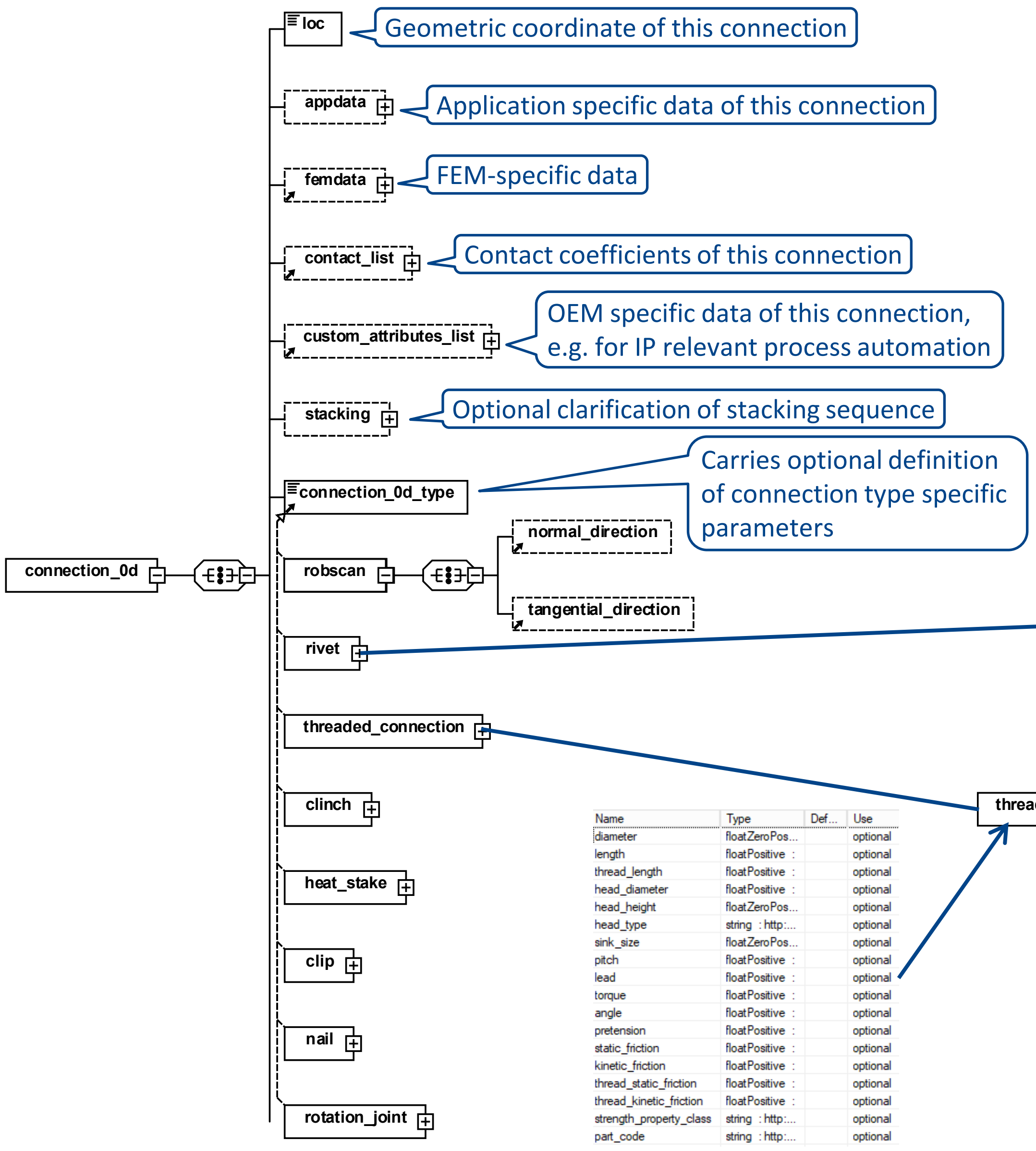
Structure of χMCF 3.1.1 XSD – High Level

- High level structure of χMCF is straight forward and easy to understand.

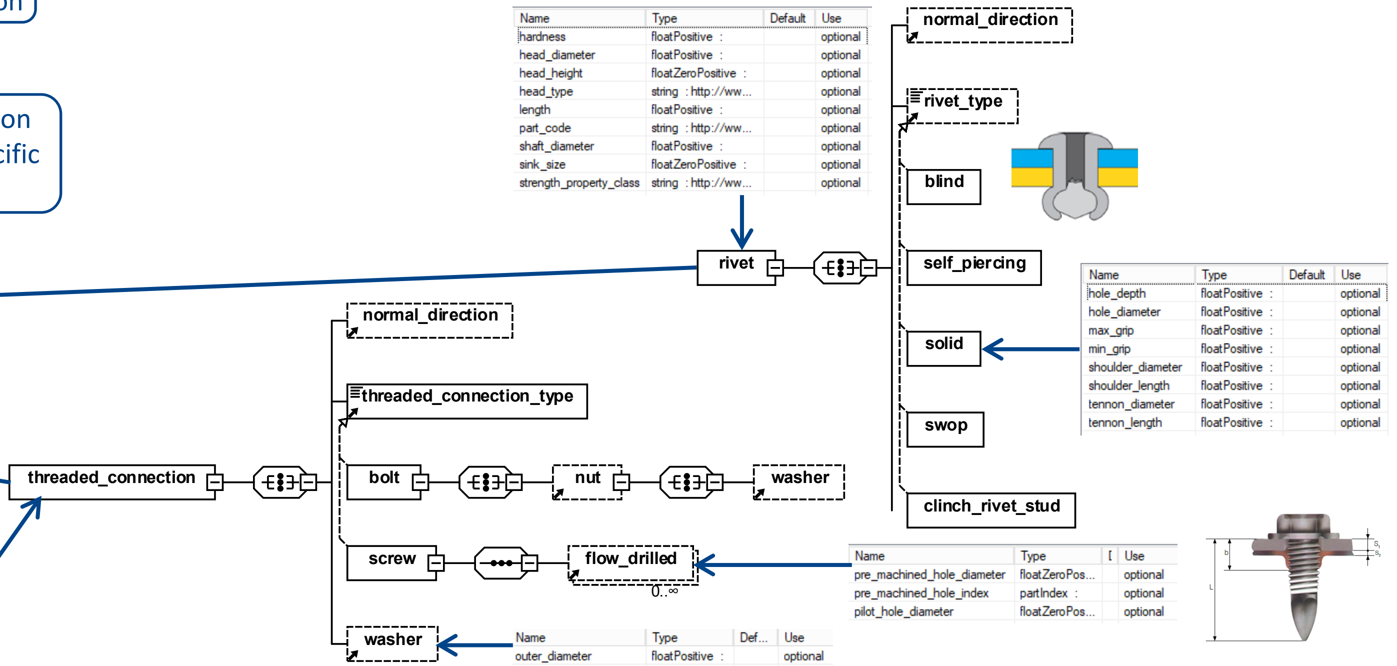


Structure of χMCF 3.1.1 XSD – Point Connections

- For point connections, 8 types are supported. Each with its own, additional parameters.
- Some have sub-types and sub-elements, e.g. <threaded_connection/>.



Name	Type	Def...	Use
diameter	floatZeroPos...		optional
length	floatPositive :		optional
thread_length	floatPositive :		optional
head_diameter	floatPositive :		optional
head_height	floatZeroPos...		optional
head_type	string : http:...		optional
sink_size	floatZeroPos...		optional
pitch	floatPositive :		optional
lead	floatPositive :		optional
torque	floatPositive :		optional
angle	floatPositive :		optional
pretension	floatPositive :		optional
static_friction	floatPositive :		optional
kinetic_friction	floatPositive :		optional
thread_static_friction	floatPositive :		optional
thread_kinetic_friction	floatPositive :		optional
strength_property_class	string : http:...		optional
part_code	string : http:...		optional



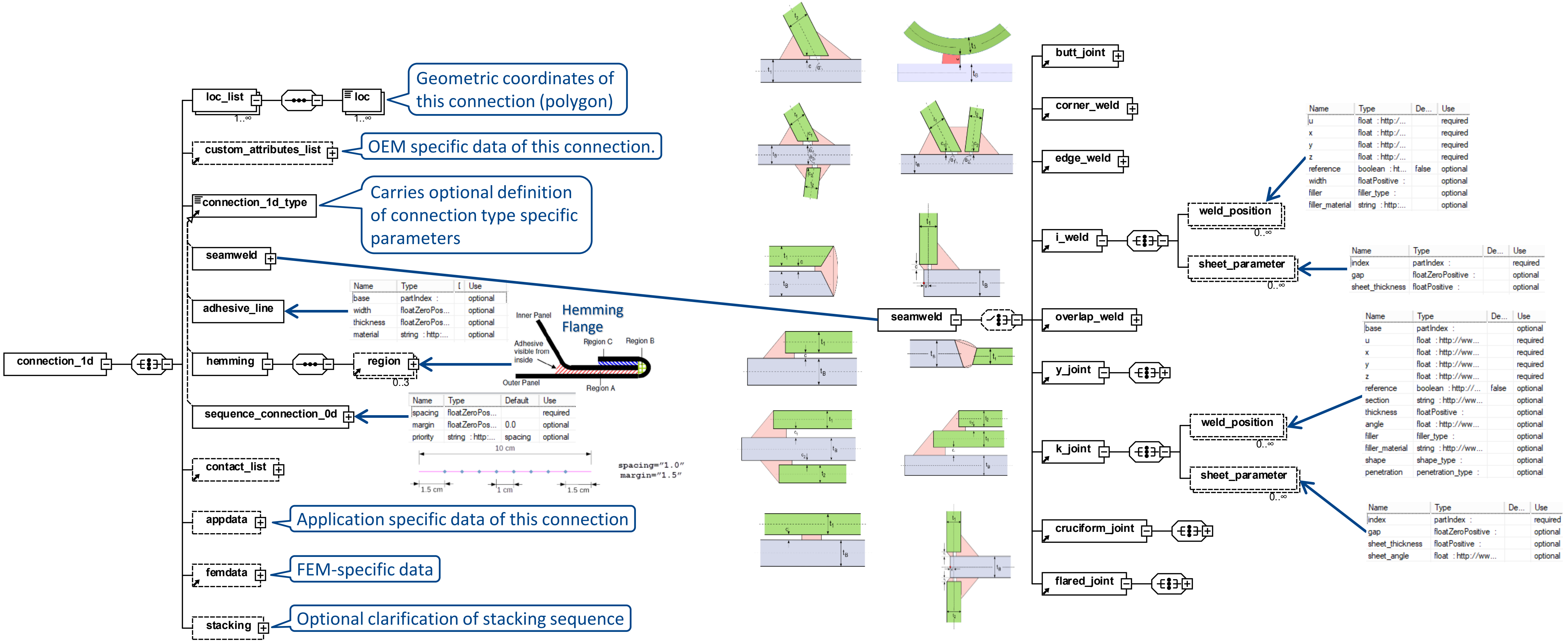
Name	Type	Default	Use
hardness	floatPositive :		optional
head_diameter	floatPositive :		optional
head_height	floatZeroPositive :		optional
head_type	string : http://ww...		optional
length	floatPositive :		optional
part_code	string : http://ww...		optional
shaft_diameter	floatPositive :		optional
sink_size	floatZeroPositive :		optional
strength_property_class	string : http://ww...		optional

Name	Type	Default	Use
hole_depth	floatPositive :		optional
hole_diameter	floatPositive :		optional
max_grip	floatPositive :		optional
min_grip	floatPositive :		optional
shoulder_diameter	floatPositive :		optional
shoulder_length	floatPositive :		optional
tennon_diameter	floatPositive :		optional
tennon_length	floatPositive :		optional

Name	Type	[	Use
pre_machined_hole_diameter	floatZeroPos...		optional
pre_machined_hole_index	partIndex :		optional
pilot_hole_diameter	floatZeroPos...		optional

Name	Type	Def...	Use
outer_diameter	floatPositive :		optional
inner_diameter	floatPositive :		optional
thickness	floatPositive :		optional
attached	boolean : ht...	false	optional
static_friction	floatPositive :		optional
kinetic_friction	floatPositive :		optional
strength_property_class	string : http:...		optional
part_code	string : http:...		optional

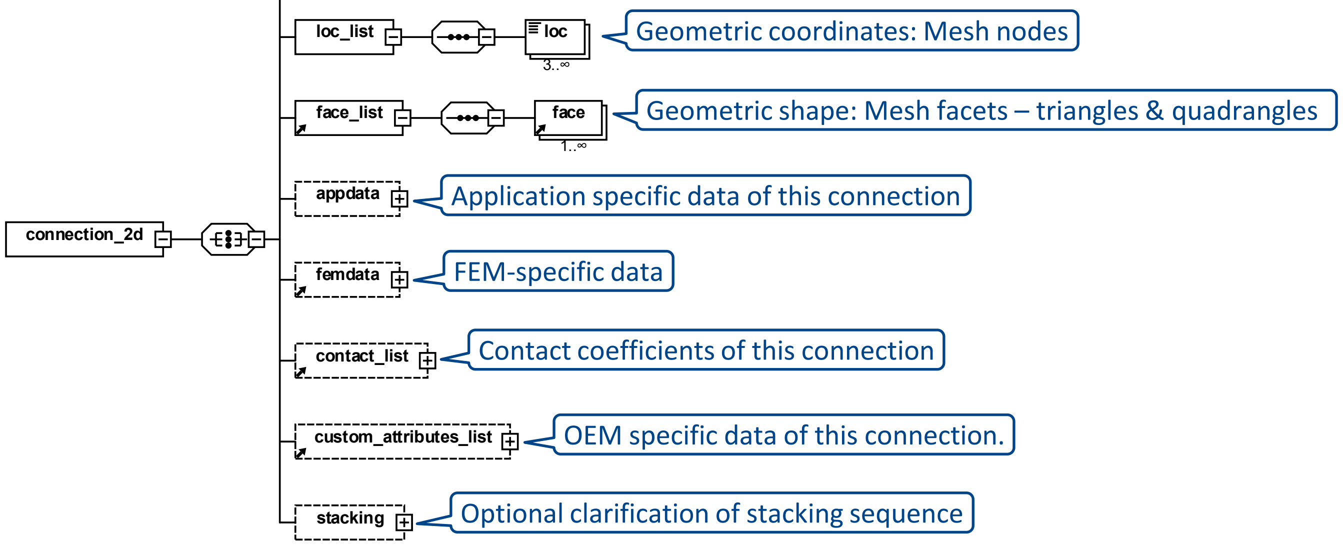
Structure of χMCF 3.1.1 XSD – Curve Connections



optional

Structure of χMCF 3.1.1 XSD – Surface Connections

- For surface connections,
  only one type is defined: adhesive gluing.



Name	Type	De...	Use
base	partIndex :		optional
thickness	floatZeroPositive :		optional
material	string : http://ww...		optional